

Intimate Relationships Weaken Female Competition: Evidence from Phase-amplitude Coupling and Event-Related Potentials

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Abstract—Competition, an essential component of social interaction, frequently occurs in daily life, and the impact of intimate relationships on women's competition has not yet been revealed. In this study, the visual target paradigm was used to explore the neural mechanisms underlying the regulation of female competitiveness by intimate relationships using event-related potential (ERP) data, time–frequency analysis, and brain functional connectivity. The research results indicate that, the P1, the N4, and the LPP were sensitive to the impact of intimate relationships on competition. Compared to competition between unfamiliar opposite-gender dyads, the average amplitudes of the N4 and LPP were higher in the late stage of competition between romantic partners. Compared to competition with strangers of the opposite gender, alpha band power was significantly higher when female individuals competed with their romantic partners. In addition, there was a positive correlation between the synchronization of activity in the frontal, parietal, and right temporal lobes of a female's brain and their degree of female engagement in competition. When a female individuals focused on the competition, activity synchronization was higher. The results indicate that competition with unfamiliar individuals of the opposite gender can make female focus on the competitive task, causing synchronous activation of corresponding brain regions. When competing with a romantic partner, women's focus decreases, their willingness to compete decreases, and the synchrony of brain functional connectivity decreases. This study suggests that intimate relationship weakens women's competitiveness, which is of significant importance for understanding high-quality intimate relationship and promoting the development of healthy competition. © 2023 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: competition, intimate relationships, ERP, time-frequency analysis, brain functional connectivity.

INTRODUCTION

Competition is one of the ways to promote individual growth (Chuang et al., 2015). In competitions, one or more individuals take some actions to achieve a goal by opposing another individual or group of the same species with the same goal; this situation is common in human communities or groups (Salvador and Costa, 2009). Previous research has suggested that male competition tends to be more focused on physical or career-related competition, often displaying dominance through strength, skills, and achievements (Rubenstein, 2012). Meanwhile, female competition is typically characterized by social competition, which involves gaining advantages or achieving goals through social relationships, appearance, charm, and other means. People often contact and communication with others. When faced with different

choices, there will inevitably be disagreements, which means that individuals have to decide whether to compete or compromise; romantic relationships are no exception (Minson and Dorison, 2022). Intimacy is defined as “feelings of closeness, connectedness, and bondedness in a close relationship” (Sternberg, 1986). Furthermore, within romantic relationships, intimacy is comprised of various components: commitment, interdependence, emotional intimacy, and physical intimacy (Blieszner and De Vries, 2001). The continuation and development of intimate relationships require management and maintenance. Benign competition can promote the growth of both individuals. In contrast, negative competition may passively frustrate individuals and even harm romantic partners, destroying the emotional bond between individuals in intimate relationships (Lee et al., 2021).

Previous studies have shown that women have relatively higher awareness of and influence over interpersonal communication processes (Acitelli, 1992). They tend to be more enthusiastic and compliant than men, while men tend to be more dominant and indifferent than women (Suh et al., 2004). In addition, research has

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Abbreviations: CFNs, canonical functional brain networks; DAN, dorsal attention; DMN, default mode networks; ERP, event-related potential; FPN, frontoparietal.

shown that gender differences in peer relationships are related to different behavioural and emotional outcomes (Alarcón et al., 2020). For example, men tend to socialize and engage in competitive interactions within a larger group of peers, while women engage in longer and more intimate binary interactions. That is, there are large gender differences in the interactive activities of individuals in different relationships, such as the closest and most intimate relationship between romantic partners. In Erikson's eight-stage theory of personality development, there is an overview of the development stages of intimacy and loneliness in early adulthood. When individuals appropriately handle their intimate relationships, they can progress to the next stage of personality development. Although classic theories enhanced past understanding, previous theories have mainly focused on men, and some researchers have questioned whether these results represent all people (Hodgson and Fischer, 1979). Social competition processes and intimate relationships seem relatively straightforward in men, but corresponding phenomena in women, who are equally likely to encounter these scenarios, have received less attention. Thus, this study focused on women and investigated how intimate relationships affect female competitiveness and the underlying cognitive and neural mechanisms.

With advances in neurophysiological techniques, researchers have conducted many studies on competition to explore the neural mechanism underlying competition and the associated neural substrates. Previous studies have shown that competition is related to the N1, the P1, the P2, the P3, the N4, and the LPP (Carretié et al., 2001; Beyens et al., 2015; Chuang et al., 2015; Zhou et al., 2015; Dyson et al., 2020). In previous studies, it has been found that the N1 component is closely related to early stimulus processing and attention allocation in individuals, while the late component P3 can measure individuals' attention and motivational responses to stimuli (Dyson et al., 2020). The P1 is mainly generated in the visual occipital area (Natale et al., 2006); this event-related potential is related to an individual's attention to the competition and varies according to the degree and intensity of the individual's attention. Greater attention to competitive stimuli leads to shorter P1 latencies and higher average amplitudes (Luck et al., 1990; Zanto et al., 2010). The P2 component is associated with individuals' attention allocation to stimuli, and after experiencing a failure event, individuals may exhibit greater attention allocation to negative stimuli (Carretié et al., 2001). Conversely, when individuals win a competition, the resulting positive emotions can also lead to larger average amplitudes of the P2 component. The N4 reflects the subjective emotional response to stimuli. The N4 amplitude is larger when the competition is meaningful to the individual or when the individual's emotions are influenced by the competition (such as in intimate relationships) (Imbir et al., 2023). Finally, the LPP occurs during late cognitive reappraisal, representing conscious emotional arousal in the late stage of individual processing. Beyens et al. (2015) found that factual feedback during competition is reflected in individuals' LPP (late positive potential), and when individuals receive an apology from

a competitor, their willingness to compete decreases, accompanied by an increase in the average amplitude of the LPP, indicating that the LPP reflects individuals' cognitive neural state of emotion and motivation in the late stage of competition.

In addition, brain activity in the theta and alpha frequency bands also plays a vital role in competition according to time–frequency analyses (Song et al., 2022; Wang et al., 2023). The theta frequency band mainly reflects an individual's cognitive and intellectual resources and is a marker of cognitive load (Sauseng et al., 2010). When individuals are more invested in competitive tasks, brain activation in the theta frequency band increases. The alpha frequency band reflects individual alertness. It is related to the level of cognitive processing, such as visuospatial attention and cognitive resources during motor processing. In competitions, this band reflects the degree of engagement in competitive tasks (Song et al., 2022). Recent meta-analyses have suggested that social cognitive processes, such as inferring the intentions of others, seem to involve balanced interactions of functional integration and isolation between several canonical functional brain networks (CFNs) (Schurz et al., 2020). Shaw et al. (2023) argued that there is high demand for flexible cognition and adaptive behaviour in competitive interactions, in which individuals must respond to the actions of familiar players; these abilities are related to the connectivity among the dorsal attention (DAN), frontoparietal (FPN), and default mode networks (DMN). Putri et al. (2022) compared brain connectivity between winners and losers of competitive games. They found that the winners successfully controlled activity at the Pz electrode, while the losers exhibited greater activation of the left frontal cortex related to cooperation. The information flow from the winners to the losers was greater than that from the losers to the winners, which indicated that the winners monitored their opponents more closely than the losers. This study explored the influence of intimate relationships on competition from the perspective of individual brain connectivity and inter-brain connectivity to clarify the psychological processes involved in competition and provide support for relevant research on competition.

In this study, the effect of intimate relationships on female competition and the underlying neural mechanisms were explored using event-related potential data, time–frequency analysis, and brain functional connectivity. The hypotheses were as follows. (A) Competition between romantic partners would increase the average amplitude of late event-related potential (ERP) components, such as the N4 and the LPP. (B) Competition involves allocating attentional resources, and competition between romantic partners is influenced by their intimate relationship, resulting in reduced attention. Compared to competition between strangers, competition between romantic partners is more likely to result in higher alpha band activity and lower theta band activity in women. (C) Because individuals in romantic relationships are more likely to cooperate rather than compete, the neural synchrony of women in the romantic group may decrease. When competing with

unfamiliar individuals, winning the competition is the only goal, which may lead to enhanced neural synchrony.

EXPERIMENTAL PROCEDURES

Participants

We used G^* Power (Faul et al., 2007) to determine the sample size needed. According to the calculations (parameters: effect size $d = 0.8$, $\alpha = 0.5$, $1 - \beta = 0.8$, independent-sample t test), the required sample size was 52. A total of 52 female college students were recruited (age: 18–25 years, $M = 20.10$ years old, $SD = 3.0$); 26 women were placed in the group competing with romantic partners (hereafter, romantic group), and 26 were placed in the group competing with strangers (hereafter, stranger group). To reduce the impact of confounding variables, women in the stranger group were also in a romantic relationship and had not met their opposite-gender interaction partner before the experiment. Previous studies have showed that women's responses to social pressures may be affected by the menstrual cycle (Kirschbaum et al., 1999; Duchesne and Pruessner, 2013; Albert et al., 2015; Banis and Lorist, 2017). Thus, female participants in this study were in the same phase of the menstrual cycle to control for the potential impact of the female menstrual cycle on competition with romantic partners. All participants were right-handed, had normal or corrected to normal vision/hearing, did not report a history of neurological/psychiatric disorders or drug abuse, and were unaware of the experiment's purpose. Romantic partners reported being intensely in love (duration of "being in love": $M = 7.20$ months, range = 2–36 months). Participants were informed before the experiment that different rewards would be given based on the outcome of their respective competitions. The winners would receive a reward of 20 yuan, while the losers would only receive 10 yuan. Each participant signed an informed consent form before the experiment and was paid 10–20 yuan for participating. The study was approved by the medical ethics committee of the local university and followed the tenets of the Declaration of Helsinki.

Tasks and procedures

The experiment consisted of a visual cue task (Cui et al., 2012; Cheng et al., 2015). Subjects were instructed to respond to the stimuli as soon as possible. A real interactive competition scenario was adopted, as this scenario was closer to real-life situations than competing against a computer in an imagined competition. Moreover, the presence of real competitors can ensure focus on completing the experiment. The participants were seated 50 cm in front of a computer display. Women were seated on the left side, and men were seated on the right side. As this study mainly focused on comparisons between conditions, the impact of seat position was not considered. To avoid confounding effects of the stimuli or environmental differences between conditions, the same stimuli and the same presentation order were used in both conditions,

which differed only in the individual's intention to perform the task.

Fig. 1 shows the sequence of events during the task. First, the experimental requirements were described and corresponding instructions were presented on the screen. Before the formal experiment, a practice session including five trials was conducted. This allowed participants to become familiar with the task procedures. Each trial began with a blank screen presented for 2 s, followed by the appearance of a grey ring in the centre of the screen. This grey ring was the cue and had a random duration between 0.6 and 1.5 s. After this cue period, the target stimulus was presented. The target stimulus was a green circle with a grey border. Dyad members were instructed to press a button on their respective gamepad when the target stimulus appeared.

The target stimulus remained on the screen until both participants responded. After the participants responded, a feedback screen was presented for 4 s. During the experiment, the participants were required to press the keys as quickly as possible.

In the task, the original score of both participants was 100 points. Subsequently, three points were added to the winner's total score, while three points were subtracted from the loser's total score. After both participants pressed the button, a feedback screen appeared that displayed the winner and loser on the current trial and their respective points (those gained/lost and the cumulative points). Then, the next trial began after a black-screen interval of 2000 ms. The formal experiment included three blocks; each block contained 20 trials, for a total of 60 trials.

Transparency and openness

The sample size calculation and exclusion criteria for data were described above. This study was approved by the medical ethics committee of the local university and followed all tenets of the Declaration of Helsinki. All participants provided written informed consent. The data are not publicly available, but code and anonymized data can be provided upon reasonable request by the corresponding author. The design and analyses of this study were not preregistered.

Data acquisition

A 64-channel electroencephalography (EEG) recording system produced by Neuroscan Company was used to record individual EEG signals. The 64-channel SynAmps 2 recorded EEG signals and was connected to an amplifier through a headbox. According to the 10–20 international EEG acquisition system, the EEG data of individuals was collected using a 64-lead Ag/AgCl electrode cap, and data were continuously collected throughout the experiment. Both vertical and horizontal electromyography data (VEOG and HEOG data) were recorded during the experiment using bipolar recording. VEOG electrodes were placed 1 cm above and 1 cm below the median left orbit, respectively, and HEOG electrodes were placed at the lateral canthi of the eyes. The EEG data were passed through an online bandpass

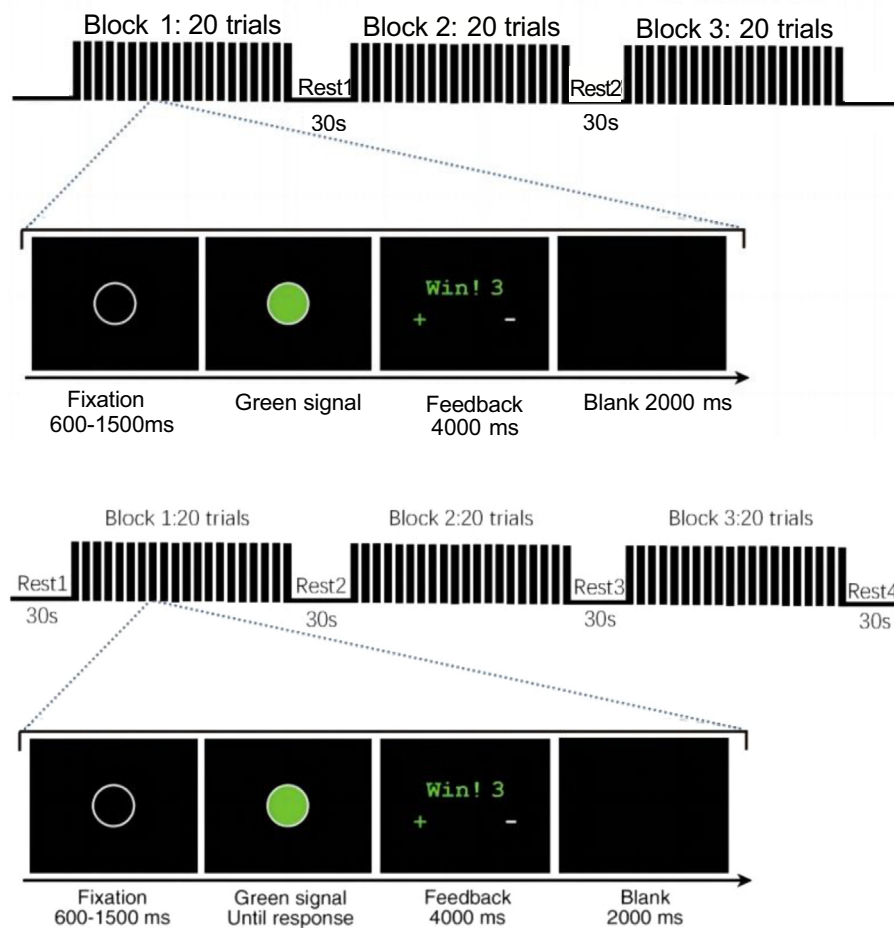


Fig. 1. Experimental design.

filter from 0.05 to 100 Hz with a sampling rate of 1,000 Hz. The impedance between all electrodes and the scalp was kept below 5 k Ω .

EEG data were preprocessed using MATLAB R2018b with the EEGLAB toolbox (Makeig et al., 2004). Data pre-processing involved locating electrodes, removing unused channels (HEOG, VEOG) using the whole brain average as a reference, performing 1 Hz high-pass filtering and 45 Hz low-pass filtering, downsampling to 500 Hz, setting the time of stimulus onset as the zero point, segmenting the data, performing baseline correction using the time before stimulus onset as the baseline, performing independent component analysis, and removing wave amplitudes exceeding $\pm 100 \mu\text{V}$ (artefacts). EEG data were examined in the window of -2000 to 1000 ms, with the stimulus onset time set as the zero point. ERP data segments from -200 to 800 ms were analysed, and time–frequency analysis, traceability and functional connectivity analysis were conducted with EEG data from -2000 to 1000 ms, with stimulus onset time set as the zero point. During EEG recording, the individual's reaction time was recorded with E-prime 3.0 (as the latency from the initial presentation of the stimulus to the subject's response), and SPSS 23.0 was used for statistical analysis.

Data analysis

Behavioural performance. The average response time (RT) of female individuals, RT of male individuals, and the outcomes of the winning trials were recorded. To quantify female performance, we calculated the percentage of winning trials (PWT) (Cui et al., 2012; Cheng et al., 2015). First, the K-S test was used to check the normality of variables; the RT and PWT data were normally distributed. The magnitude of the effect in an independent-sample t test was calculated using Cohen's d ; values > 0.2 indicated a small effect, values > 0.5 indicated a medium effect, and values ≥ 0.8 indicated a large effect (Cohen, 1988). The α level was set at 0.05 for all tests. Finally, Spearman correlation analyses were performed to explore the relationship between ERP data and behaviour.

ERP and time-frequency data analysis. In this study, the P1 (time window: 120–160 ms after stimulus onset), N4 (330–370 ms), and LPP (500–750 ms) were examined (Pires et al., 2014; Clark et al., 2019; Šoškić et al., 2022), and independent-sample t tests comparing data from 62 electrodes among conditions were carried out

in MATLAB R2018b by the data-driven method. The theta frequency band and alpha frequency band were selected for time–frequency analysis. An independent-sample t test was performed using a data-driven method for 64 electrodes among the conditions.

Traceability and functional connectivity analysis. To further determine the brain regions associated with competition, a distributed source model was constructed with FieldTrip minimum norm estimation (MNE) to enable source localization for the preprocessed scalp EEG data. Multiple comparisons were corrected with a cluster-based permutation test method, with a significance threshold of 0.05, and 1,000 permutation tests were conducted.

The phase locking value (PLV) was selected as a measure of functional connectivity derived from individual EEG data (Lachaux et al., 1999). In short, this method involves calculating the phase difference between activity at all electrode pairs in a time window and then evaluating the stability of the phase difference across all experiments (Burgess, 2013). To avoid volumetric conduction effects, a current source density (CSD) analysis was performed on the preprocessed data before calculating the PLV. Furthermore, the time–frequency matrix of all segmented stored

phase information was obtained using a short-time Fourier transform. To generate the phase matrix of each pair of electrodes in an area, φ_i and φ_j were set as the positive vector of the signal phase of the two electrodes i and j in the time window t and at frequency f . The phase difference between two electrodes was calculated as follows:

$$\varphi_{ij} = \varphi_i \cdot \varphi_j \times (\varphi_j \times \text{the complex conjugate of } \varphi_i)$$

For each time–frequency point, the consistency of the phase differences of all segments at that time–frequency point were calculated to obtain the PLV of each time–frequency point as follows:

$$PLV_{ij} = \text{abs}(1/N \times \sum \phi_{ij})$$

where N is the number of tests. The PLV was calculated for each pair of electrodes (i, j) and ranges from 0 to 1, where 0 indicates that the two signals are unsynchronized, and 1 indicates perfect synchronization. The baseline was set at -800 to -200 ms before stimulus onset for baseline correction. Based on the literature and the spectrum generated in this study, -200 to 1000 ms was selected as the period of interest. The theta band ($3 \sim 7$ Hz) and alpha band ($8 \sim 12$ Hz) were defined as the frequency bands of interest, and an independent-sample t test and FDR correction were performed. The α level was set at 0.05 for all tests. The same sensory stimuli and stimuli presentation order were used to avoid confounding effects of stimulus or context between conditions. The only difference between conditions was the intention with which the participants performed the task. All data are presented as the mean $\pm$ SD.

Correlation analysis. To understand the relationship between the time domain, frequency domain, and functional connectivity related to competitive behaviour, we examined the behavioural index, average ERP amplitude and average frequency band power. Significant results were further explored with Spearman correlation analysis and the PLV to identify the cognitive mechanisms of individual competition.

RESULTS

Behavioural performance

Independent-sample t tests were conducted on women's RT and PWT during competition, and the results are shown in Fig. 2. The PWT of the romantic group ($M \pm SD = 53.27 \pm 9.54$) was significantly greater than that in the stranger group ($M \pm SD = 46.47 \pm 12.73$) ($t(50) = 2.18, p = 0.034$, Cohen's $d = 0.60$). There was no significant difference in RT.

For males, there was no significant difference in RT between the two groups. However, in the romantic group, males' RT ($M \pm SD = 275.72 \pm 42.83$) was significantly higher ($t(50) = 2.89, p = 0.006$, Cohen's $d = 0.82$) than females' RT ($M \pm SD = 244.49 \pm 34.70$). In the stranger group, males' RT ($M \pm SD = 280.20 \pm 42.24$) was significantly higher

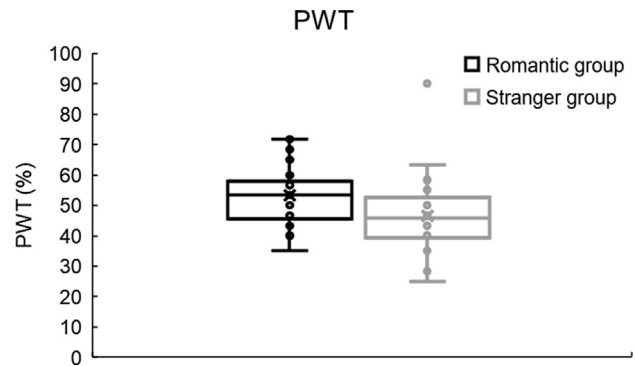


Fig. 2. Percentage of winning trials (PWT) indicates the likelihood that the woman would win the competition. The “box” depicting the median and the 25th and 75th quartiles and the “whisker” showing the 5th and 95th percentile.

($t(50) = 2.15, p = 0.036$, Cohen's $d = 0.61$) than females' RT ($M \pm SD = 257.86 \pm 31.93$) (Fig. 3).

ERP results

The results showed that the electrodes with significant differences in P1 amplitude according to group were AF4, FT8, T8, CPZ, CP2, TP8, P3, P1, PZ, and P2. The electrodes with significant differences in N4 amplitude according to group were P7, P5, P3, PO7, and PO5. The electrodes with significant differences in LPP amplitude according to group were P8, PO6, PO8, and O10.

Further independent-sample t tests of the average amplitudes at these electrodes showed that the average P1 amplitude in women in the stranger group ($M \pm SD = 0.46 \pm 0.66$) was significantly greater than that in women in the romantic group ($M \pm SD = 0.05 \pm 0.58$) ($t(50) = -2.36, p = 0.022$, Cohen's $d = -0.65$). The average N4 amplitude of the romantic group ($M \pm SD = 4.09 \pm 2.94$) was significantly greater than that of the stranger group ($M \pm SD = 2.44 \pm 2.22$) ($t(50) = 2.30, p = 0.026$, Cohen's $d = 0.65$). The average LPP amplitude of the romantic group ($M \pm SD = -0.02 \pm 3.46$) was significantly greater than that of the stranger group ($M \pm SD = -2.50 \pm 2.77$) ($t(50) = 2.86, p = 0.006$, Cohen's $d = 0.81$) (Fig. 4).

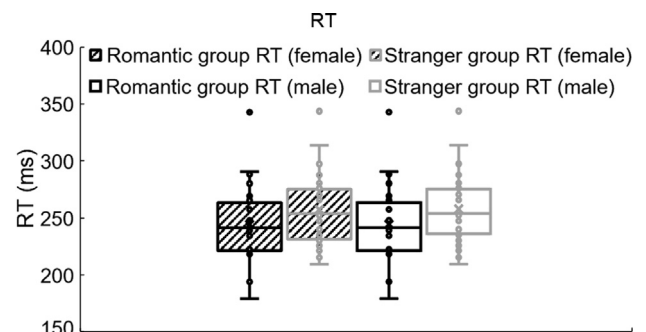


Fig. 3. Average reaction time of individuals competing for buttons. The “box” depicting the median and the 25th and 75th quartiles and the “whisker” showing the 5th and 95th percentile.

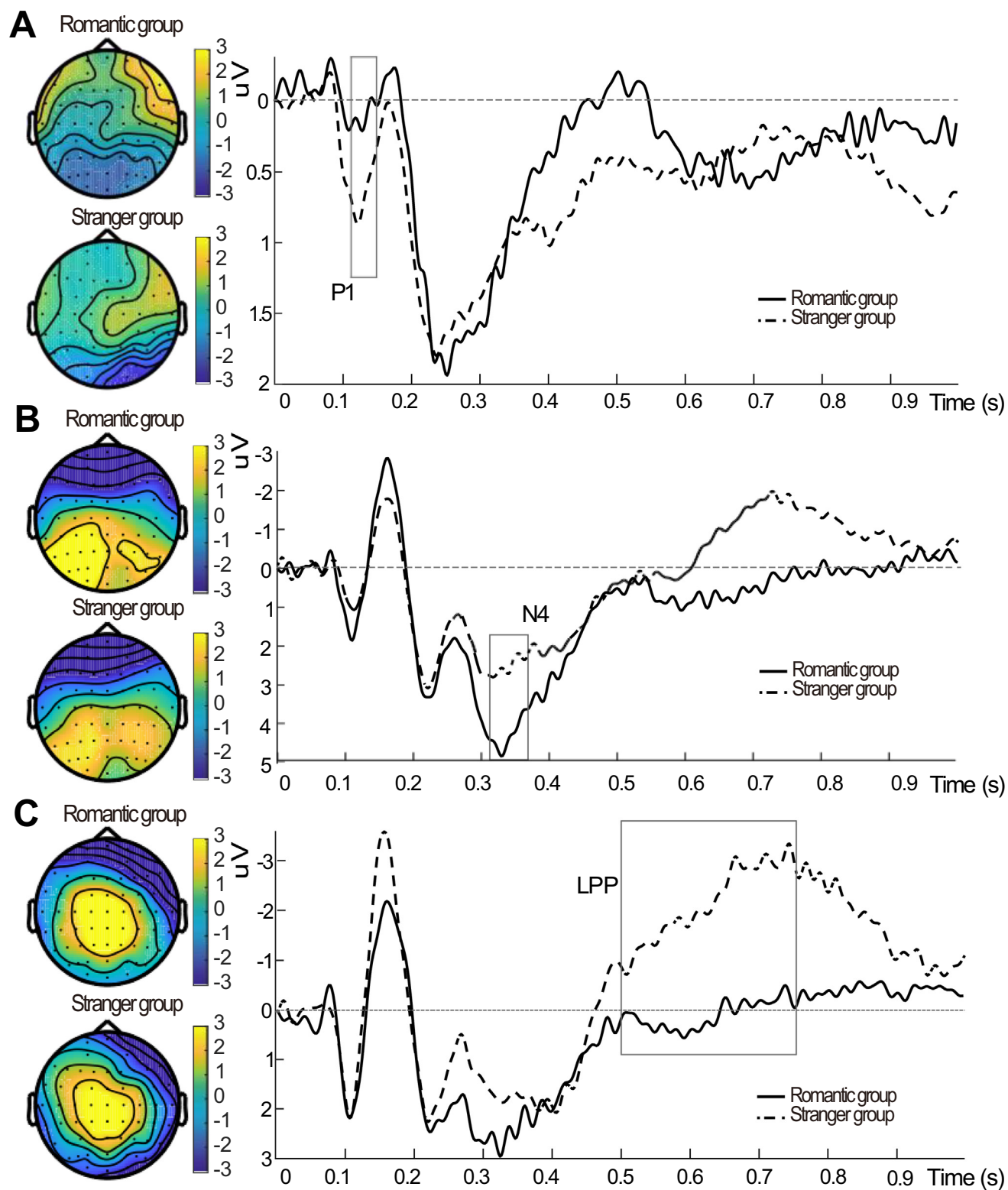


Fig. 4. Brain topographic map of the components and the average waveform corresponding to the electrodes. (A) Brain topographic map and waveform of the P1 in each group. (B) Brain topographic map and waveform of the N4 in each group. (C) Brain topographic map and the LPP waveforms of each group.

Time-frequency analysis results

The results showed a significant difference in theta band activity among the F6, F8, CP5, P3, P1, PZ, and POZ electrodes in the 0–400 ms period; the F8 electrodes in the 400–600 ms period; and the F2, PO3, PO4, and P1 electrodes in the 600–800 ms period. There were significant differences in alpha band activity among the FT8, P5, and O2 electrodes in the 0–200 ms period; the F7, F5, F1, F4, F6, F8, and FC6 electrodes in the 200–400 ms period; and the F7, F6, F8, FT7, FC6, FT8, T7, T8, TP7, and P2 electrodes in the 600–800 ms period.

Furthermore, independent-sample *t* tests were performed on the average power of these electrodes, and the results are shown in Fig. 5. In the alpha frequency band, there were significant differences in power in the 200–400 ms ($t(50) = 2.31$, $p = 0.025$, Cohen's $d = 0.64$), 400–600 ms ($t(50) = 2.62$, $p = 0.012$, Cohen's $d = 0.73$), and 600–800 ms periods ($t(50) = 2.33$, $p = 0.024$, Cohen's $d = 0.65$). Alpha power was greater in the romantic group ($M \pm SD = 0.85 \pm 1.74$, 0.90 ± 1.89 , and 0.99 ± 2.18 , respectively) than in the stranger group ($M \pm SD = -0.34 \pm 1.97$, -0.61 ± 2.25 , and -0.62 ± 1.75). There were no significant group differences in theta power.

EEG synchronization results

The results showed that during the 0–200 ms period, there was a significant difference in the PLV in the theta band ($t(50) = 2.09$, $p = 0.041$, Cohen's $d = 0.59$). The synchronization between the FC2 and FT8 electrodes in women in the romantic group ($M \pm SD = 0.31 \pm 0.12$) was significantly higher than that in women in the stranger group ($M \pm SD = 0.24 \pm 0.12$). At 800–1,000 ms, the PLV results in the theta frequency band showed that the stranger group ($M \pm SD = 0.31 \pm 0.16$) had significantly higher brain synchronization at the F7 and FP1 electrodes than the romantic group ($M \pm SD = 0.23 \pm 0.13$) ($t(50) = -2.06$, $p = 0.044$, Cohen's $d = -0.58$).

In addition, the PLV in the alpha frequency band in the 0–200 ms period was significantly higher in the stranger group ($M \pm SD = 0.32 \pm 0.10$) than in the romantic group ($M \pm SD = 0.23 \pm 0.08$) ($t(50) = -3.75$, $p < 0.001$, Cohen's $d = -1.06$) (Fig. 6).

Further analysis of the EEG signal showed that the results mainly involved the right temporal, frontal, and parietal regions, as shown in Fig. 7.

Correlation analysis results

First, a Spearman correlation analysis was performed on the PLV and PWT. The results showed that at 800–1,000 ms, the PLV in the theta band between the F7 and FP1 electrodes was significantly negatively correlated with the PWT ($r = -0.29$, $p = 0.036$); that is, the higher the probability of women winning was, the weaker the functional synchronization of the F7 and FP1 electrodes.

The correlation between the average amplitude of ERP components and the PLV showed that the PLV in the theta frequency band between the F7 and FP1 electrodes in the 800–1,000 ms period was significantly positively correlated with the P1 amplitude ($r = 0.32$, $p = 0.023$). In addition, at 0–200 ms, the PLV of the alpha frequency band between the CP4 and O10 electrodes was also significantly positively correlated with the P1 amplitude ($r = 0.36$, $p = 0.009$). In other words, the higher the average P1 amplitude was, the higher the synchronization of the F7–FP1 and CP4–O10 pairs.

Finally, the results of the correlation between time–frequency power values and the PLV showed that at 0–200 ms, the PLV in the alpha frequency band between the CP4 and O10 electrodes was significantly negatively correlated with alpha power in the 200–800 ms period (200–400 ms: $r = -0.35$, $p = 0.011$; 400–600 ms: $r = -0.38$, $p = 0.006$; and 600–800 ms: $r = -0.35$, $p = 0.011$). The lower the alpha power was, the higher the synchronization between the CP4 and O10 electrodes (Fig. 8).

Discussion

In this study, ERP, time–frequency analysis, and functional connectivity techniques were applied to investigate the neural mechanisms underlying the impact of intimate relationships on female competition. The results show that intimate relationships weakened female competition and the evidences also supported the research hypotheses of the competition effects between romantic relationship on women brain activity.

The behavioural results indicated that women had higher winning rates than males when they competed with their romantic partners, but there was no significant difference in RT between the females in two groups. This suggests that females do not show a significant difference in competitiveness when competing against individuals from different groups. The higher winning rate among females in the romantic group may be ascribed to the behavior and competitive strategies of the male competitors. The RT results for males showed that males in both groups exhibited weaker competitive abilities compared to females, with males in the romantic group displaying even weaker abilities. This finding contradicts the traditional notion that males have an advantage over females in terms of physical movement and competition speed (Cashdan, 1998; Kivikangas et al., 2014). Previous research has reported that in various cultures, people perform mate-retention behaviours that either provide a benefit to their partner or inflict a cost on themselves to maintain a long-term relationship, prevent infidelity and avoid relationship dissolution (Buss, 1988; Buss and Shackelford, 1997; Barbaro et al., 2015). A partner retention strategy is a way to maintain long-term and stable intimate relationships. Women had a higher winning rate when competing with their romantic partners, which may be the result of men adopting spouse retention strategies to maintain their romantic relationships when competing with their romantic partners to prevent the partners from leaving or being poached by other competitors (Atari et al.,

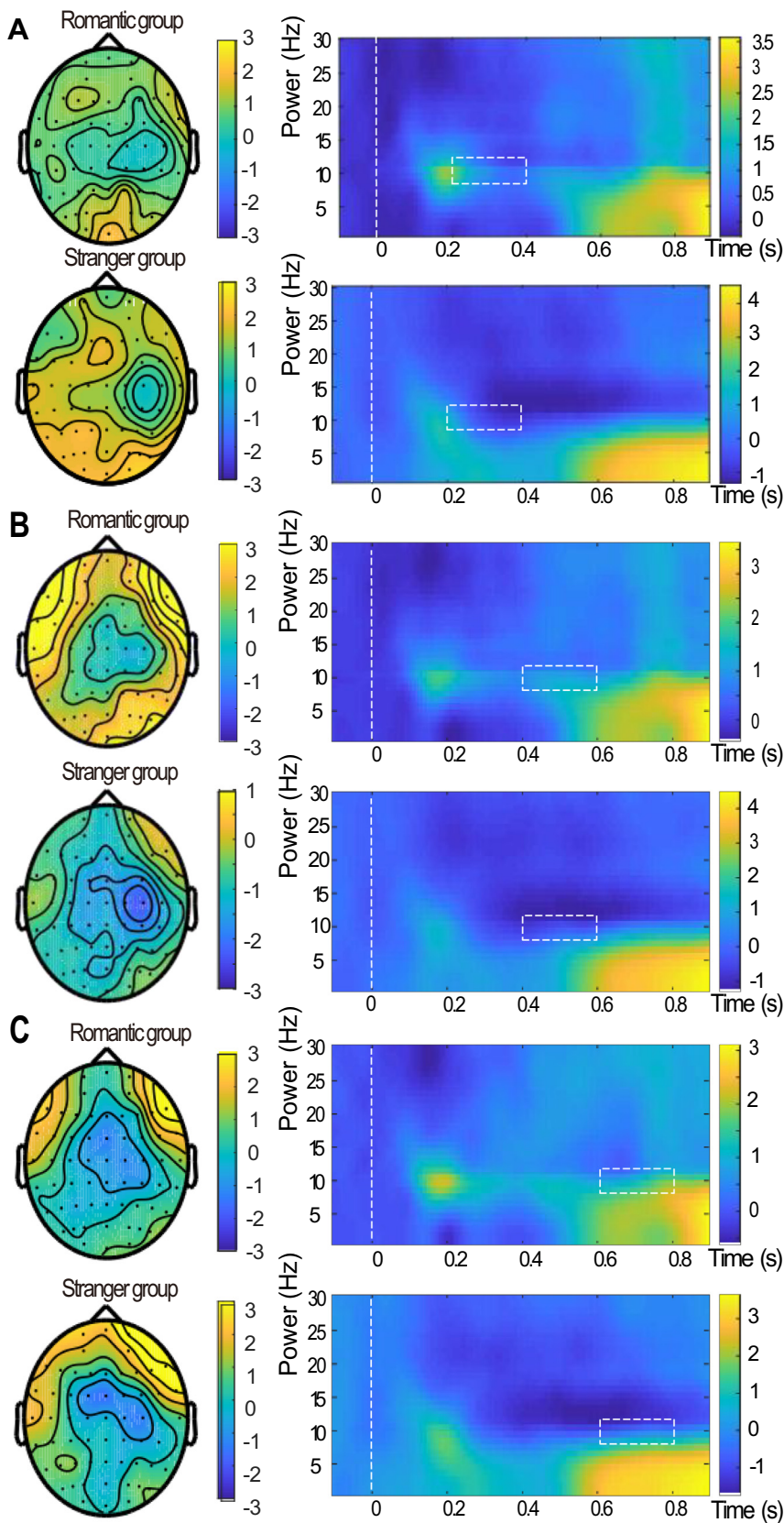


Fig. 5. Brain topographic map and grand average time–frequency distributions of the alpha band activity. (A) Topographic map and grand average time–frequency distributions in the 200–400 ms period. (B) Topographic map and grand average time–frequency distributions in the 400–600 ms period. (C) Topographic map and grand average time–frequency distributions in the 600–800 ms period.

2017), which is consistent with sexual selection theory regarding mate choice (Eastwick et al., 2017).

The ERP results showed that in the parietal lobe, the average P1 amplitude was significantly lower in the romantic group than in the stranger group, and the average amplitudes of the N4 and LPP were significantly higher. Previous studies have shown that the P1 is closely related to an individual's attention. The more an individual concentrates on a task, the higher the average P1 amplitude, and the better an individual can cope with competitive stimuli. Moreover, during the early period, in the P1 time window, individuals are mainly not affected by emotions (Kulke et al., 2022). In this study, we hypothesized that when competing with unfamiliar individuals, women would be more focused on the competition, without substantial distractions. In the experiments, we found that when females competed with unfamiliar individuals, they had higher P1 amplitudes, which is consistent with the hypothesis. Brain activity is a response driven by stimulus information (Xin et al., 2023). When women competed with their romantic partner, they had lower P1 amplitudes, which means that women were not as focused on the task when they competed with an intimate partner.

In contrast, the LPP reflects processing of results and emotional perception (Schindler and Bublatzky, 2020), and the N4 reflects the further processing of meaningful stimuli (Manfredi et al., 2018). In this study, compared to the stranger group, the romantic group may have exhibited more in-depth cognitive processing, which means that women believed that their competitive interaction with their romantic partner was more meaningful. Notably, regardless of whether this meaning was related to the competition task itself, it elicited the N4 and LPP (Russell et al., 2019). Regarding the behavioural results, during the competition process, the PWT was higher in the romantic group than in the stranger group. Research has shown that stronger

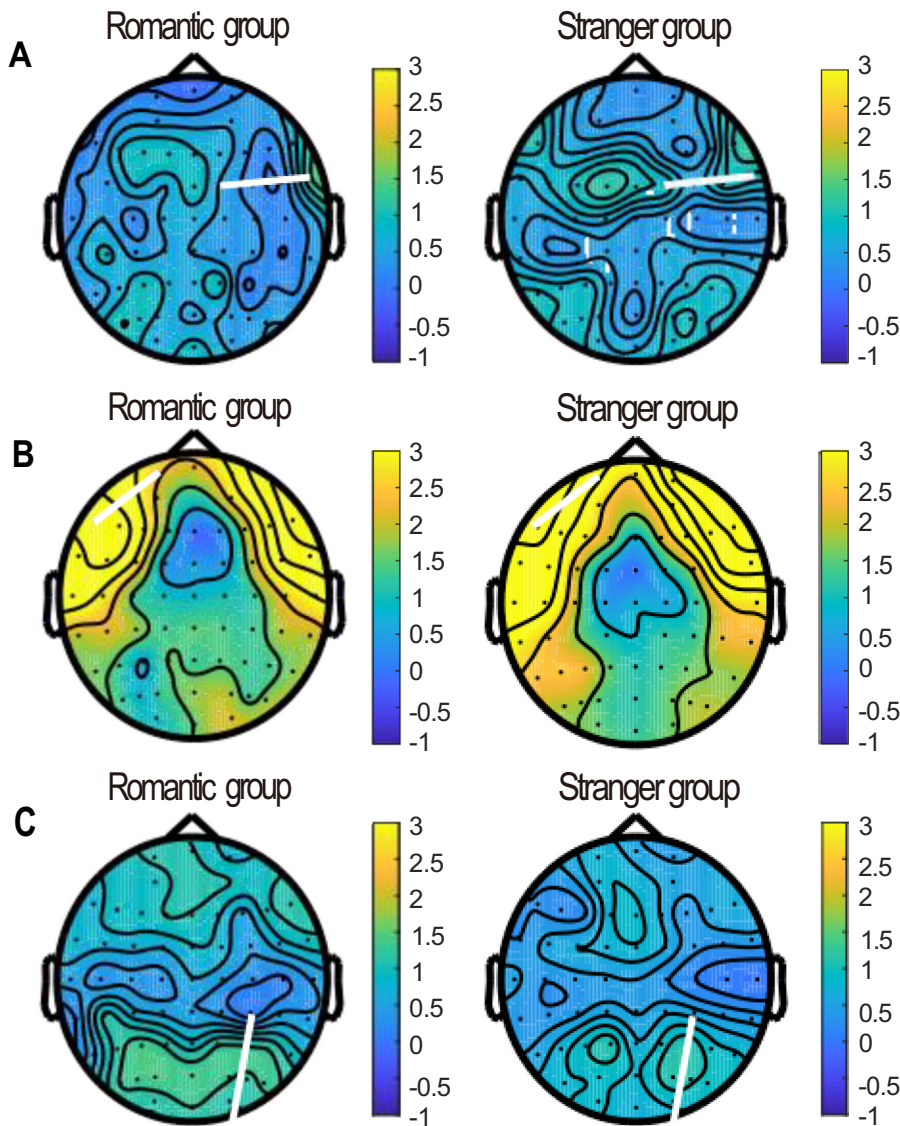


Fig. 6. Difference diagram of the functional connectivity between two groups of electrodes in each frequency band. (A) At 0–200 ms, the FC2 and FT8 electrodes were functionally connected and distributed at 3–5 Hz. (B) At 800–1,000 ms, the F7 and FP1 electrodes were functionally connected and distributed at 3–7 Hz. (C) At 0–200 ms, the CP4 and O10 electrodes were functionally connected and distributed at 8–12 Hz.

neurobiological responses to environmental rewards may alleviate emotional responses to stressors when individuals face typical or competitive pressures (Fassett-Carman et al., 2023). In this study, individuals who have won before, similar to obtaining environmental rewards, may experience reduced pressure to compete and increased self-motivation (Moturu and Lent, 2023), thus feeling satisfied with their abilities and grateful for the humility of their partners. During deep processing of stimuli and perception of emotions, larger N4 and LPP amplitudes were generated. However, this stimulus processing and emotional perception may be due to the presence of their romantic partner rather than the competitive task itself.

In the romantic group, women exhibited significantly greater alpha frequency band activity than in the stranger group. Based on previous studies on alpha

frequency band activity in the frontal lobe, this may indicate that women exhibited less depth of cognitive processing when interacting with their romantic partners. In the presence of their (familiar) intimate partner, women tend to be more relaxed, less alert, and less engaged in competitive tasks (Scagliusi et al., 2016). This has also been demonstrated using functional connectivity data.

In the 200–800 ms period after stimulus onset, the stranger group exhibited greater synchronization of theta band activity between the F7 and FP1 electrodes than the romantic group. Decety et al. (2004) performed a functional magnetic resonance imaging (fMRI) study and reported that competitive behaviour activates the frontoparietal network, which supports executive functions. In addition, Mo et al. (2023) also confirmed that functional connectivity between the frontoparietal lobes is related to cognitive control in individuals that allows them to cope with interference when completing tasks; these findings can help researchers to predict the response speed of individuals and are consistent with those of the present study. Regarding EEG dynamics, numerous studies have reported that theta activity is associated with executive functions such as working memory (Fell and Axmacher, 2011; Clouter et al., 2017) and cognitive control (Cavanagh and Frank, 2014; Cooper et al., 2019). Synchronization of theta frequency band activity in the frontoparietal lobe can be considered a crucial indicator of

engaging in competitive behaviour. The higher this synchronization is, the more individuals are focused on the competitive tasks. In addition, in the early stage of competition, the synchrony of theta frequency band activity between the FC2 and FT8 electrodes in the romantic group was higher than that in the stranger group. Moreover, this synchrony gradually weakened over time. This may indicate that active attentional control decreased over time in the romantic group (Liegel et al., 2022). Based on the behavioral results, it can be observed that females in the romantic group did not exert additional competitive effort to increase their button-pressing speed, yet they had a higher winning rate in the competition. This suggests that they no longer required excessive consumption of cognitive attentional resources, leading to a

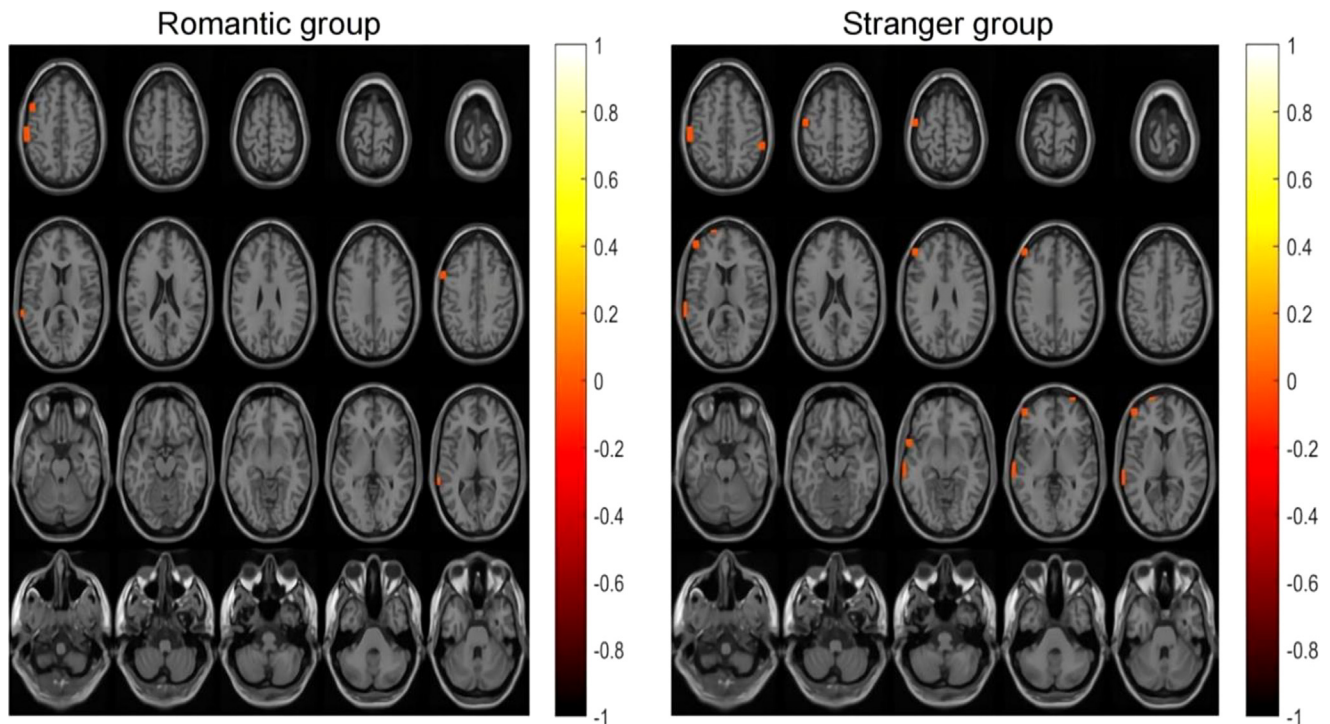


Fig. 7. Functional magnetic resonance images of the mean posterior brain activity in women competing with their romantic partners (left) and competing with strangers (right) during 0–1000 ms of stimulation. Overlaid colour indicates the significantly activated brain regions.

reduction in synchronized brain activity in the theta frequency band (Margraf et al., 2023). On the other hand, men's concession also weakens women's competitive motivation. Competition is an interactive process, and when one party chooses not to compete, the other party's competitive efforts and perceived pressure will also be affected, thereby weakening the competitive motivation of female individuals (Catalán et al., 2023).

The PLV and behavioural results showed that the higher an individual's PWT was, the lower the synchronization of activity between the F7 and FP1 electrodes. Previous studies have shown that functional connectivity between the frontal and parietal lobes is related to cognitive control during individual task completion, and stronger connectivity reflects greater cognitive effort (Mo et al., 2023). In this study, compared to the stranger group, the romantic group exhibited lower synchronization of activity in the frontal and parietal lobes, which may indicate that women exerted less cognitive effort when competing with their romantic partner. Regarding the PLV in the alpha frequency band, alpha frequency band activity decreased as the task progressed. The lower the alpha power was, the higher the individual's brain synchronization. In other words, as the competitive task progressed, individuals' concentration on the task increased regardless of group (Song et al., 2022). Notably, compared to the stranger group, the romantic group exhibited lower brain synchronization and higher alpha power. This means that when competing with an intimate partner, women's attention and alertness to the competitive task was lower. That is, intimate relationships may hinder competition in women, consistent with our hypothesis.

Nevertheless, this study has some limitations. First, due to venue constraints, the experiment was conducted on university campuses, and the participants were college or graduate students. The research results thus may only explain social interactions and competition among romantic partners on university campuses and may not be generalizable to romantic partners of other ages or in other environments. Future research should recruit subjects of different ages and use different scenarios to further explore the impact of intimacy on competition. Second, previous studies have shown that alpha and theta activity in the resting state can impact an individual's task performance (Putri et al., 2022). However, this study only explored brain activity in the competitive task and did not collect baseline EEG data from the resting state. In future research, individual data during the resting and task states could be collected to comprehensively illustrate and predict individual competitive activities. Third, due to the ambiguous boundary between friendship and love among friends of the opposite gender, even when using supervisor reports or measurement scales, individuals could still conceal or falsely report being in a relationship, which may have biased the results (Szymkow and Frankowska, 2022). Therefore, this study did not recruit familiar participants for the stranger group, such as friends who had known each other for similar lengths of time. Future research is needed to refine the experimental design of the control group, further distinguishing brain activity solely caused by familiarity without the influence of intimate relationships.

In summary, this study examined the influence of intimate relationships on competition through brain synchrony, ERP and time–frequency data collected from

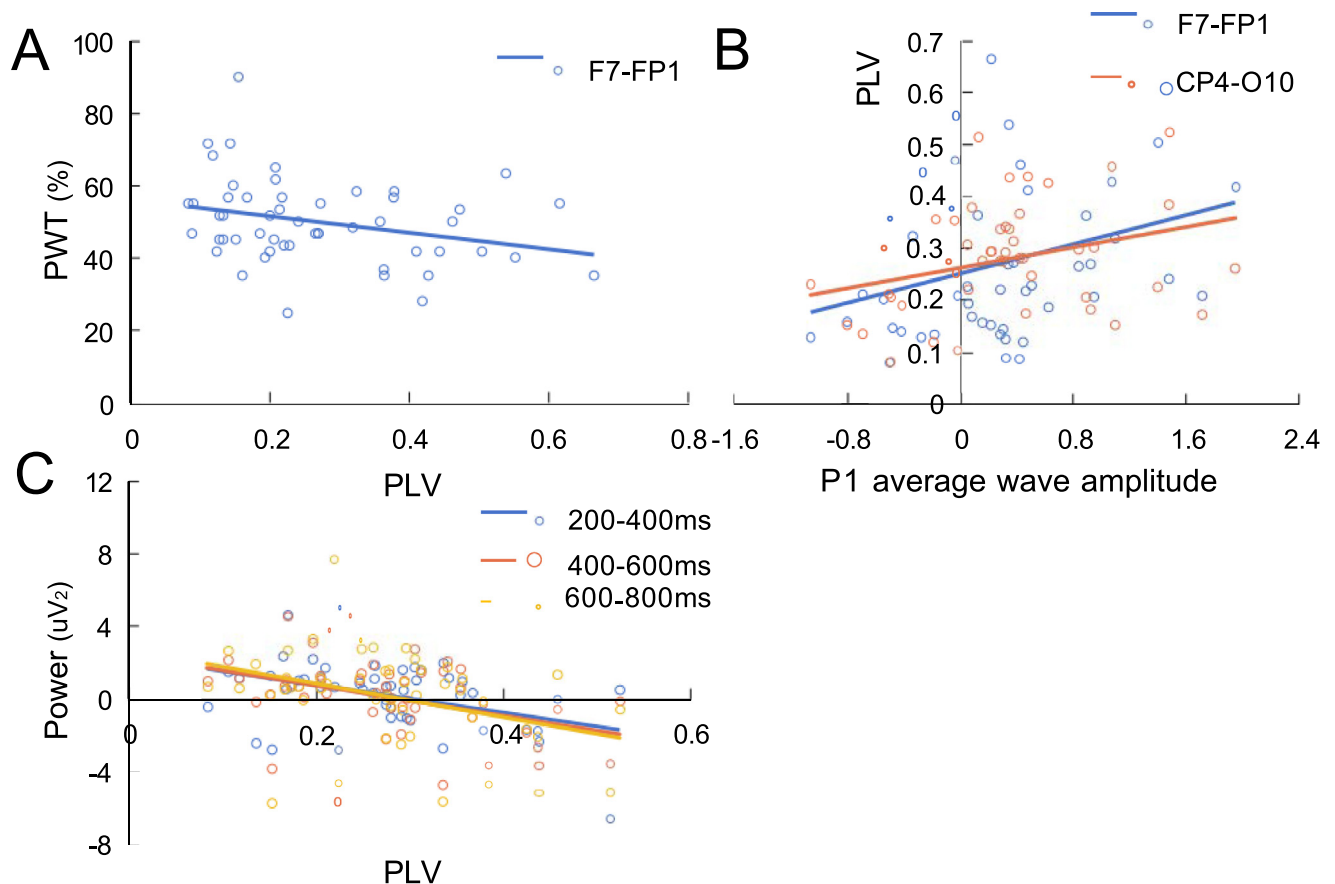


Fig. 8. (A) At 800–1,000 ms, the PLV in the theta band between the F7 and FP1 electrodes was related to the PWT. (B) Correlation between the P1 amplitude and the PLV in the F7–FP1 and CP4–O10 pairs. (C) The PLV in the alpha frequency band between the CP4 and O10 electrodes at 0–200 ms was correlated with alpha power at 200–800 ms.

individuals using EEG technology. Our results indicated that intimate relationships affect women's competitive attention and cognitive control, with reduced brain activity observed when competing with romantic partners and increased brain activity observed when competing with strangers. Specifically, differences were observed in the P1, the N4, the LPP, and the alpha and theta frequency bands. This study investigates the impact of intimate relationships on female competition from the perspective of cognitive neuroscience, which is highly important for cultivating high-quality intimate relationships and promoting the development of healthy social competition.

AUTHORS' CONTRIBUTION

Y.L. and S.J. designed the study; Y.M. and S.J. conducted the experiments, collected data and analyzed the data; Q.W., Y.G., L.A., L.Y. and H.W. contributed to conceptualization, software, investigation and writing—original draft; Y.L. and S.J. wrote the paper.

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DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

CONFLICT OF INTEREST

The authors declared that they had no conflict of interest with respect to their authorship or the publication of this article.

REFERENCES

- Acitelli LK (1992) Gender differences in relationship awareness and marital satisfaction among young married couples. *Pers Soc Psychol Bull* 18:102–110.
- Albert K, Pruessner J, Newhouse P (2015) Estradiol levels modulate brain activity and negative responses to psychosocial stress across the menstrual cycle. *Psychoneuroendocrinology* 59:14–24.
- Alarcón G, Morgan JK, Allen NB, Sheeber L, Silk JS, Forbes EE (2020) Adolescent gender differences in neural reactivity to a friend's positive affect and real-world positive experiences in social contexts. *Dev Cogn Neurosci* 43 100779.

- Atari M, Barbaro N, Sela Y, Shackelford TK, Chegeni R (2017) The big five personality dimensions and mate retention behaviors in Iran. *Pers Individ Differ* 104:286–290.
- Banis S, Lorist MM (2017) The combined effects of the menstrual cycle phase and acute stress on reward-related processing. *Biol Psychol* 125:130–145.
- Barbaro N, Pham MN, Shackelford TK, Zeigler-Hill V (2015) Insecure romantic attachment dimensions and frequency of mate retention behaviors. *Pers Relat* 23(3):605–618.
- Beyens U, Yu H, Han T, Zhang L, Zhou X (2015) The strength of a remorseful heart: psychological and neural basis of how apology emolliates reactive aggression and promotes forgiveness. *Front Psychol* 6:1611.
- Blieszner R, De Vries B (2001) Introduction perspectives on intimacy. *Generations* 25(2):7–8.
- Burgess AP (2013) On the interpretation of synchronization in EEG hyperscanning studies: a cautionary note. *Front Hum Neurosci* 7:881–898.
- Buss DM (1988) From vigilance to violence: tactics of mate retention in American undergraduates. *Ethol Sociobiol* 9:291–317.
- Buss DM, Shackelford TK (1997) From vigilance to violence: mate retention tactics in married couples. *J Pers Soc Psychol* 72:346–361.
- Catalán JM, Blanco-Ivorra A, García-Pérez JV, Vales Y, Martínez-Pascual D, Ezquerro S, Garrote A, Costa T, Lledó LD, García-Aracil N (2023) Patients' physiological reactions to competitive rehabilitation therapies assisted by robotic devices. *J Neuroeng Rehabil* 20(1):41–55.
- Carretié L, Mercado F, Tapia M, Hinojosa JA (2001) Emotion, attention, and the 'negativity bias', studied through event-related potentials. *Int J Psychophysiol: J Int Org Psychophysiol* 41(1):75–85.
- Cashdan E (1998) Are men more competitive than women? *Br J Soc Psychol* 37(Pt 2):213–229.
- Cavanagh JF, Frank MJ (2014) Frontal Theta as a mechanism for cognitive control. *Trends Cogn Sci* 18(8):414–421.
- Cheng X, Li X, Hu Y (2015) Synchronous brain activity during cooperative exchange depends on the gender of partner: a fNIRS-based hyperscanning study. *Hum Brain Mapp* 36(6):2039–2048.
- Chuang LY, Huang CJ, Hung TM (2015) Effects of attentional training on visual attention to emotional stimuli in archers: a preliminary investigation. *Int J Psychophysiol: J Int Org Psychophysiol* 98(3 Pt 1):448–454.
- Clark AP, Bontemps AP, Batky BD, Watts EK, Salekin RT (2019) Psychopathy and neurodynamic brain functioning: a review of EEG research. *Neurosci Biobehav Rev* 103:352–373.
- Clouter A, Shapiro KL, Hanslmayr S (2017) Theta phase synchronization is the glue that binds human associative memory. *Curr Biol* 27(20):3143–3148.e6.
- Cohen J (1988) *Statistical Power Analysis for the Behavioral Sciences*. second ed. Routledge.
- Cooper PS, Karayanidis F, McKewen M, McLellan-Hall S, Wong ASW, Skippen P, Cavanagh JF (2019) Frontal Theta predicts specific cognitive control-induced behavioural changes beyond general reaction time slowing. *Neuroimage* 189:130–140.
- Cui X, Bryant DM, Reiss AL (2012) NIRS-based hyperscanning reveals increased interpersonal coherence in superior frontal cortex during cooperation. *Neuroimage* 59(3):2430–2437.
- Decety J, Jackson PL, Sommerville JA, Chaminade T, Meltzoff AN (2004) The neural bases of cooperation and competition: an fMRI investigation. *Neuroimage* 23:744–751.
- Duchesne A, Pruessner JC (2013) Association between subjective and cortisol stress response depends on the menstrual cycle phase. *Psychoneuroendocrinology* 38(12):3155–3159.
- Dyson BJ, Steward BA, Meneghetti T, Forder L (2020) Behavioural and neural limits in competitive decision making: the roles of outcome, opponency and observation. *Biol Psychol* 149 107778.
- Eastwick PW, Harden KP, Shukusky JA, Morgan TA, Joel S (2017) Consistency and inconsistency among romantic partners over time. *J Pers Soc Psychol* 112(6):838–859.
- Fassett-Carman AN, Moser AD, Ruzic L, Neilson C, Jones J, Barnes-Horowitz S, Schneck CD, Kaiser RH (2023) Amygdala and nucleus accumbens activation during reward anticipation moderates the association between life stressor frequency and depressive symptoms. *J Affect Disord* 330:309–318.
- Faul F, Erdfelder E, Lang AG, Buchner A (2007) G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav Res Methods* 39(2):175–191.
- Fell J, Axmacher N (2011) The role of phase synchronization in memory processes. *Nat Rev Neuroscience* 12(2):105–118.
- Hodgson JW, Fischer JL (1979) Sex differences in identity and intimacy development in college youth. *J Youth Adolesc* 8(1):37–50.
- Imbir KK, Duda-Góławska J, Wielgopalan A, Sobieszek A, Pastwa M, Zygierewicz J (2023) The role of subjective significance, valence and arousal in the explicit processing of emotion-laden words. *PeerJ* 11:e14583.
- Kirschbaum C, Kudielka BM, Gaab J, Schommer NC, Hellhammer DH (1999) Impact of gender, menstrual cycle phase, and oral contraceptives on the activity of the hypothalamus-pituitary-adrenal axis. *Psychosom Med* 61(2):154–162.
- Kivikangas JM, Kätsyri J, Järvelä S, Ravaja N (2014) Gender differences in emotional responses to cooperative and competitive game play. *PLoS One* 9(7):e100318.
- Kulke L, Brümmer L, Pooresmaeli A, Schacht A (2022) Visual competition attenuates emotion effects during overt attention shifts. *Psychophysiology* 59(11):e14087.
- Lachaux JP, Rodriguez E, Martinerie J, Varela FJ (1999) Measuring phase synchrony in brain signals. *Hum Brain Mapp* 8(4):194–208.
- Lee AE, Rivera C, Montes SB, Jimenez-Zambrano A, Nacht A, Bolanos A, Asturias E, Berman S, Heinrichs G, Harrison MS (2021) Identifying prevalence and risk factors for intimate partner violence in pregnant women in rural Guatemala. *J women's Health Dev* 4(4):113–122.
- Liegel N, Schneider D, Wascher E, Arnau S (2022) Task prioritization modulates Alpha, Theta and beta EEG dynamics reflecting proactive cognitive control. *Sci Rep* 12(1):15072.
- Luck SJ, Heinze HJ, Mangun GR, Hillyard SA (1990) Visual event-related potentials index focused attention within bilateral stimulus arrays. II. Functional dissociation of P1 and N1 components. *Electroencephalogr Clin Neurophysiol* 75(6):528–542.
- Makeig S, Delorme A, Westerfield M, Jung TP, Townsend J, Courchesne E, Sejnowski TJ (2004) Electroencephalographic brain dynamics following manually responded visual targets. *PLoS Biol* 2(6):e176–e192.
- Manfredi M, Cohn N, De Araújo Andreoli M, Boggio PS (2018) Listening beyond seeing: event-related potentials to audiovisual processing in visual narrative. *Brain Lang* 185:1–8.
- Margraf L, Krause D, Weigelt M (2023) Frontal theta reveals further information about neural valence-dependent processing of augmented feedback in extensive motor practice – a secondary analysis. *Eur J Neurosci* 57(8):1297–1316.
- Minson JC, Dorison CA (2022) Toward a psychology of attitude conflict. *Curr Opin Psychol* 43:182–188.
- Mo Z, Grennan G, Kulkarni A, Ramanathan D, Balasubramani PP, Mishra J (2023) Parietal Alpha underlies slower cognitive responses during interference processing in adolescents. *Behav Brain Res* 443 114356.
- Moturu BP, Lent RW (2023) Self-assertive efficacy and workplace advocacy behavior: a social cognitive analysis. *J Couns Psychol* 70(1):41–51.
- Natale E, Marzi CA, Girelli M, Pavone EF, Pollmann S (2006) ERP and fMRI correlates of endogenous and exogenous focusing of visual-spatial attention. *Eur J Neurosci* 23(9):2511–2521.
- Pires L, Leitão J, Guerrini C, Simões MR (2014) Event-related brain potentials in the study of inhibition: cognitive control, source localization and age-related modulations. *Neuropsychol Rev* 24(4):461–490.
- Putri F, Susnoschi Luca I, Garcia Pedro JA, Ding H, Vučković A (2022) Winners and losers in brain computer interface competitive

- gaming: directional connectivity analysis. *J Neural Eng* 19(4). <https://doi.org/10.1088/1741-2552/ac8451>.
- Rubenstein DR (2012) Sexual and social competition: broadening perspectives by defining female roles. *Philos Trans R Soc Lond Series B, Biol Sci* 367(1600):2248–2252.
- Russell MJ, Masuda T, Hioki K, Singhal A (2019) Culture and neuroscience: how Japanese and European Canadians process social context in close and acquaintance relationships. *Soc Neurosci* 14(4):484–498.
- Sauseng P, Griesmayr B, Freunberger R, Klimesch W (2010) Control mechanisms in working memory: a possible function of EEG Theta oscillations. *Neurosci Biobehav Rev* 34(7):1015–1022.
- Salvador A, Costa R (2009) Coping with competition: neuroendocrine responses and cognitive variables. *Neurosci Biobehav Rev* 33(2):160–170.
- Scagliusi FB, da Rocha Pereira P, Unsain RF, de Moraes Sato P (2016) Eating at the table, on the couch and in bed: an exploration of different locus of commensality in the discourses of Brazilian working mothers. *Appetite* 103:80–86.
- Schindler S, Bublatzky F (2020) Attention and emotion: an integrative review of emotional face processing as a function of attention. *Cortex; J Devoted Study Nervous Syst Behav* 130:362–386.
- Schurz M, Maliske L, Kanske P (2020) Cross-network interactions in social cognition: a review of findings on task related brain activation and connectivity. *Cortex; J Devoted Study Nervous Syst Behav* 130:142–157.
- Shaw DJ, Czekóová K, Mareček R, Havlice Špiláková B, Brázdil M (2023) The interacting brain: dynamic functional connectivity among canonical brain networks dissociates cooperative from competitive social interactions. *Neuroimage* 269 119933.
- Song M, Zhao Q, Du C, Zhou C, Li R (2022) The relationship between the accuracy of curling athletes' duration judgment and delivery performance. *PeerJ* 10:e13541.
- Šoškić A, Jovanović V, Styles SJ, Kappenman ES, Ković V (2022) How to do better n400 studies: reproducibility, consistency and adherence to research standards in the existing literature. *Neuropsychol Rev* 32(3):577–600.
- Sternberg RJ (1986) A triangular theory of love. *Psychol Rev* 93:119–135.
- Suh EJ, Moskowitz DS, Fournier MA, Zuroff DC (2004) Gender and relationships: influences on agentic and communal behaviors. *Pers Relat* 11(1):41–60.
- Szymkow A, Frankowska N (2022) Moderators of sexual interest in opposite-sex friends. *Evolutionary Psychol: Int J Evolution Approaches Psychol Behav* 20(1) 14747049211068672.
- Wang K, Li Y, Liu H, Zhang T, Luo J (2023) Relationship between pistol players' psychophysiological state and shot performance: activation effect of EEG and HRV. *Scand J Med Sci Sports* 33(1):84–98.
- Xin Z, Wang F, Lu Z, Li Q (2023) Characteristics of attentional bias in meditators: an ERP study. *Int J Psychol: J Int Psychol* 58(2):143–152.
- Zanto TP, Toy B, Gazzaley A (2010) Delays in neural processing during working memory encoding in normal aging. *Neuropsychologia* 48(1):13–25.
- Zhou L, Jiang C, Wu Y, Yang Y (2015) Conveying the concept of movement in music: an event-related brain potential study. *Neuropsychologia* 77:128–136.

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